

Programming for Computational Linguistics 2025/2026

Exercise Set 11 (2026-01-22)

Debugging and Unit Tests

Once you have completed all exercises, send an email to the `progclgrader@ims.uni-stuttgart.de` with the title “submit set11”, and with your python files as attachments. You should receive a response with feedback and a grade.

Exercises are due on **2026-02-10 23:59**, but we encourage you to do them during the lab session.

Exercise 1. Download from ILIAS the file `grades_calculator.py`. It contains two functions: `calculate_average(grades)` that should calculate an average of the given grades:

- `calculate_average([])` should return `0`
- `calculate_average([1, 1])` should return `1`
- `calculate_average([1, 2, 3])` should return `2`

and `categorize_grade(average)` that should return a grade based on the given average:

- `categorize_grade(91)` should return `"A"`
- `categorize_grade(72)` should return `"C"`
- `categorize_grade(85)` should return `"B"`

Currently, both functions have bugs. Use your favourite debugging method to find and fix these bugs. When you are ready, submit `grades_calculator.py` to the grader.

Exercise 2. Download from ILIAS file `palindrome_checker.py`. The file contains a function `is_palindrome`, that returns `True` if the input is a palindrome (it ignores case and non-alphanumeric characters):

- `is_palindrome("a")` returns `True`
- `is_palindrome("abba")` returns `True`
- `is_palindrome("abbac")` returns `False`
- `is_palindrome("a!bba")` returns `True`

Create a file named `test_palindrome_checker.py` with **unit tests** for the `is_palindrome` function. The tests should verify that palindroms like `"madam"`, `"A man, a plan, a canal, Panama"`, and `"racecar"` return `True`. Verify that non-palindromic strings like `"hello"` and `"world"` return `False`. Test edge cases such as an empty string and strings with only special characters (e.g., `"!!!"`). Submit only the file `test_palindrome_checker.py`.

Exercise 3. Create a file `calculator.py` containing a class `Calculator` with the following methods:

- `add(a, b)` : Returns the sum of a and b
- `subtract(a, b)` : Returns the difference between a and b.
- `multiply(a, b)` : Returns the product of a and b.

- `divide(a, b)` : Returns the result of dividing a by b. If b is 0, raise a `ValueError` with the message `"Cannot divide by zero"`.

Create a second file `test_calculator.py` with unit tests for the `Calculator` class. The tests should verify the correctness of the `add`, `subtract`, `multiply`, and `divide` methods. Ensure that dividing by zero raises a `ValueError`. Submit both files – `calculator.py` and `test_calculator.py`.